

REVES: Redirection Enhancement Using Four-Pole Vestibular Electrode Stimulation

Seokhyun Hwang

anoldhsh@gm.gist.ac.kr

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea

YoungIn Kim

youngin3737@gm.gist.ac.kr

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea

Jieun Lee

jieunlee@gist.ac.kr

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea

SeungJun Kim*

seungjun@gist.ac.kr

Gwangju Institute of Science and Technology
Gwangju, Republic of Korea

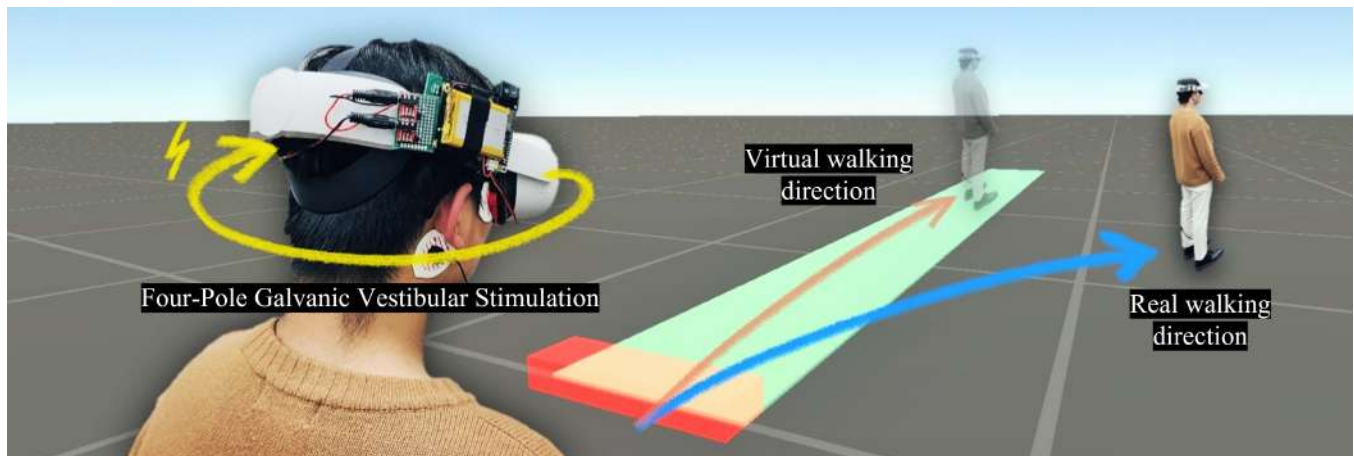


Figure 1: REVES: Redirection Enhancement Using Four-Pole Vestibular Electrode Stimulation

ABSTRACT

Redirected walking (RDW) visually manipulates the movement of the virtual environment (VE) such that the movement of the real environment (RE) and VE are no longer matched 1:1. With RDW, users can overcome the spatial constraints of RE, such as furniture, walls, and columns, and freely move the wider VE using natural gait motion. However, when the intensity of visual manipulations increases, people notice the RDW manipulation owing to visual-vestibular inconsistency, and experience discomfort like motion sickness. To address visual-vestibular inconsistency, we modulated vestibular information in various directions to match the modulation of visual information using galvanic vestibular stimulation

(GVS). We proposed a new RDW system—REVES: redirection enhancement using four-pole vestibular electrode stimulation. REVES stimulates the user's vestibular system according to various visual modulation directions using four-pole GVS based on the proposed algorithm. REVES changed the detection threshold of RDW in three manipulation cases: rotation, translation, and walking direction.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Haptic devices*.

KEYWORDS

Virtual Reality, Locomotion, Redirected Walking, Galvanic Vestibular Stimulation

*Corresponding author.

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1 INTRODUCTION

Redirected walking (RDW), a technology that continuously manipulates the virtual environments (VE) and the user's physical movement, can effectively satisfy natural walking by using natural human gait motion [20]. RDW changes the path in the real environment (RE) by manipulating the VE without the user noticing it. For example, where the user moves in a straight line in the VE, the virtual world is rotated slightly according to the user's direction, and the user's path in the RE transforms into an arc shape. However, most RDW technologies induce a change in direction by modulating the user's visual information, which causes visual-vestibular inconsistency. This visual-vestibular inconsistency causes cybersickness and reduces the presence [1, 7]. Owing to these limitations, the RDW technology has a detection threshold (DT) that, if surpassed, the user notices the change in the map and experiences motion sickness [8]. Thus, the DT determines the minimum size of real space required for free walking in a VE. Many studies have attempted to reduce the required real space [4, 11–13], but it is still insufficient for the general room and office size.

We believe that it is possible to increase the range of DT by reducing visual-vestibular inconsistency through modulating the vestibular information identical to the modulated visual information. We modulate the vestibular information through Galvanic vestibular stimulation (GVS). To provide vestibular stimulation suitable for various modulation directions of RDW, we intend to use four-pole GVS [3]. Although several RDW studies have used GVS before, it can be used only in a limited situation of looking at the ground due to the lack of freedom [24], or only non-directional noise was used [14]. We propose a novel RDW technique by utilizing the higher degree of freedom of four-pole GVS to extend the DT.

This study proposed and verified a new device and algorithm system named REVES: Redirection enhancement using four-pole vestibular electrode stimulation. REVES can modify vestibular information according to visual modulation in various directions of RDW technology, as shown in the Figure 1. The main contributions of this paper are as follows: (1) Proposal of a REVES system, which is a novel RDW technique that stimulates the vestibular system according to modulated visual information using four-pole GVS. (2) Exploration of the change in DT when various directions of vestibular stimuli are applied to users using the REVES system.

2 BACKGROUND

2.1 Redirected Walking

RDW technology is a promising solution to the challenge of moving a wider VE through the user's natural walking in a limited RE. Redirection is a technique in which the motions of the RE and the VE are no longer mapped 1:1 through manipulating the user's movement or the structure of the VE [18]. Redirection gain is a value that expresses the difference between RE and VE movements as a ratio and can be divided into rotation manipulation, translation manipulation, and walking direction manipulation [27].

All manipulation types are described in the Figure 2. The gain in rotation manipulation (Rotation gain, g_R) is expressed as the ratio of virtual rotation ($R_{virtual}$) to real rotation (R_{real}) (i.e.,

$g_R = \frac{R_{virtual}}{R_{real}}$). Similarly, the gain in translation manipulation (translation gain, g_T) is expressed as the ratio of virtual translation ($T_{virtual}$) to real translation (T_{real}) (i.e., $g_T = \frac{T_{virtual}}{T_{real}}$), and is mainly used to move longer distances than real translation within the VE. Finally, the walking direction manipulation (curvature gain, g_C) is a rotation applied continuously while the user moves forward. It adjusts the curvature of the user's real movement by slightly rotating the VE viewed by the user. The curvature gain (g_C) value is expressed as the reciprocal of the circular arc radius r (i.e., $g_C = \frac{1}{r}$). The g_C value is negative or positive when it has a left or right curvature, respectively. Redirection gains should be used within the DT so that the user does not detect them.

2.2 Detection Threshold (DT) and Point of Subjective Equality (PSE)

Several studies measure and expand DTs to overcome the spatial constraints of room-scale VR [21, 22, 26]. Steinicke et al. measured DTs using two-alternative forced-choice (2AFC) tasks in rotation, translation, and walking direction manipulations. Participants chose one of the two possible responses to questions like "Was the virtual movement smaller or greater than the physical movement?" or "Is the physical path vent left or right?" Steinicke et al. defined the point of subjective equality (PSE), in which the human subject perceives physical and virtual movements equally, based on the human subject's answers. At both sides of the PSE, the values that human subjects can discriminate between physical and virtual movements in a 75% range is set as the DT.

Other studies have extended the DT to reduce the minimum real space required for free walking in an expanded space of VE. There are vision-based ways to adjust the VE's layout or manipulate it during specific eye movements, such as saccadic [5, 29], blinking [13], and change blindness [28], but they are only available in limited circumstances or require the use of high-end eye trackers. Only a few studies have attempted to utilize auditory stimulation [22], haptic stimulation [15], and vestibular stimulation [14, 24]. The vestibular stimulation study first presented by Sra can be used only in a limited situation where the user walks while looking at the ground owing to the lack of freedom. In a recent study by Matsumoto et al., there was plenty of room for improvement by using only noisy GVS, which has no direction. The study extended the DT of curvature gain from 12% to 16% by disrupting the vestibular system using only the noisy GVS without direction and showed the possibility of improving the RDW performance using GVS. Moving further from the method that only confuses the vestibular system, we utilize the higher degree of freedom in the direction of stimulation using four-pole GVS that can be used in various directions of RDW gains.

2.3 Galvanic Vestibular Stimulation

GVS electrically stimulates the vestibular system to induce virtual head motion in a bipolar direction. Since it was discovered that passing an electric current through the mastoids on the back of both ears in humans could upset the balance and disrupt the vestibular system, GVS has been generally used for medical purposes [8, 16]. Recently, HCI researchers have applied GVS in other applications including balance games [6], interactive tools among users [25],

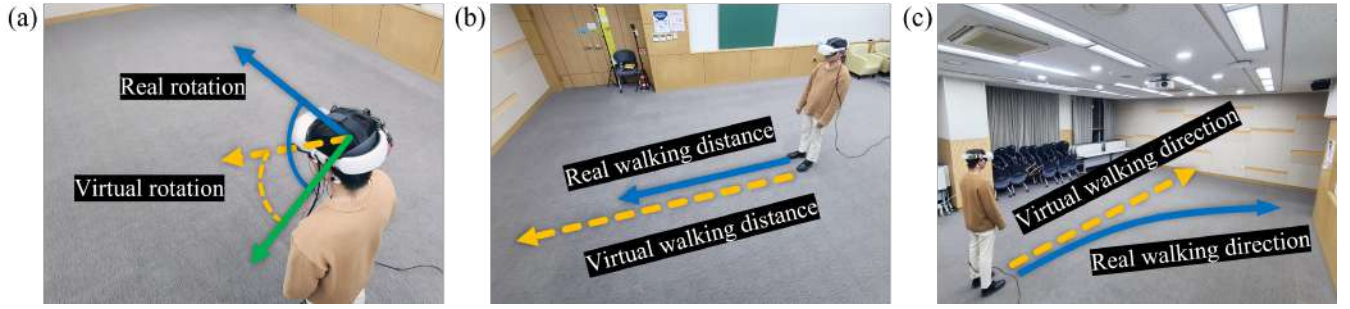


Figure 2: Images explaining types of manipulation. (a) Rotation manipulation: rotating VE more or less than RE by head-mounted display (HMD) rotations. (b) Translation manipulation: moving VE faster or slower than RE. (c) Walking direction manipulation: rotating VE while walking straight in VE results in arc move in RE.

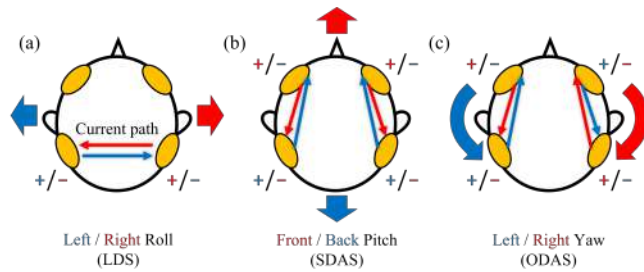


Figure 3: Current paths according to GVS signals. The yellow ellipses mark the positions of the electrodes attached on the head. The thin red and blue arrows indicate the directions of the current flow, and the large blue and red arrows indicate the direction of virtual head motion for (a) lateral directional stimulation, (b) same directional anteroposterior stimulation, and (c) opposite directional anteroposterior stimulation conditions.

and roller coaster simulations [2, 17], with higher sense of presence using directional vestibular stimulation in VR. Based on these previous studies, our 4-pole GVS device was manufactured with additional functions described in section 3.1.

In addition to the GVS using unidirectional current, noisy GVS without directionality and multi-pole GVS using multiple electrodes were introduced. The noisy GVS showed that the vestibular sensation could be improved by generating stochastic resonance through white noise current stimulation [9, 30]. The multi-pole GVS demonstrated that virtual head motion could be induced in various directions by attaching additional electrodes [3, 23]. Four-pole GVS, a type of multi-pole GVS, is a system with a high degree of freedom that can induce virtual head motion of roll, pitch, and yaw in three rotation axes. By using four-pole GVS, three different vestibular stimuli can be applied by attaching electrodes to the left temple, right temple, left mastoid, and right mastoid. As shown in Figure 3, six current directions are created in lateral directional stimulation (LDS), same directional anteroposterior stimulation (SDAS), and opposite directional anteroposterior stimulation (ODAS) by selecting the anode and cathode for each electrode. According to the current direction, the user experiences rotation in each of the three axes

directions. We selected a four-pole GVS that has a high degree of freedom for an RDW system that can be used in various situations.

3 IMPLEMENTATION

3.1 Hardware Implementation

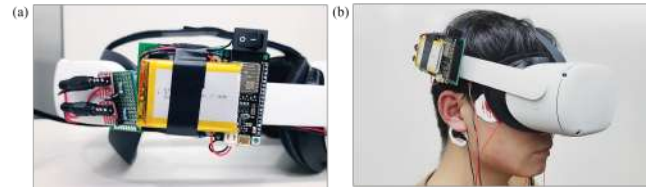


Figure 4: (a) REVES system device attached to the side of the Oculus Quest 2 elite strap. (b) REVES system attached to the participant.

The proposed REVES system device (Figure 4) remotely stimulates the vestibular system by attaching electrodes to the four positions of the left temple, right temple, left mastoid, and right mastoid, and providing six different current directions. The system generates a biphasic voltage of 15 V. It complies with the proposed safety standards of ± 2 mA maximum current limit.

The microcontroller unit (MCU) board processes the signals received from the experimental environment running in Unity. The direction of the galvanic current applied to the user is controlled through the dual H-bridge circuit of the L293D motor driver and can be adjusted through the pulse-width modulation signal. Electrical stimulation is provided as a ramp function to prevent unexpected tingling caused by electric current. The generated signal is output through a 3.5-mm connector and delivered to the user through a conductive hydrogel electrode. The integrated Inertial Measurement Unit (IMU) sensor on the board can measure the user's head orientation. Based on the user's head direction information from the IMU and RDW gain information in Unity, the device converts the stimulus direction from the global coordinate system to the head coordinate system, using the following operation algorithm. Because the entire device design is based on open-source hardware, it can be universally used by adding more electrodes and sensors.

3.2 Methodology

We developed a REVES system that modulates vestibular information to match visual information modulated by RDW technology to reduce visual-vestibular inconsistency. Our system works for rotation, translation, and walking direction manipulation, as follows:

Rotation Manipulation. We can apply two RDW techniques to the user in a rotation manipulation environment: a virtual rotation greater than the real rotation ($g_R > 1$) or a virtual rotation smaller than the real rotation ($g_R < 1$). If the user experiences a larger visual virtual rotation than the real rotation ($g_R > 1$), the vestibular system is stimulated in the same yaw rotation direction on the global coordinate system to solve the visual-vestibular inconsistency. Conversely, when a virtual rotation smaller than the real rotation ($g_R < 1$) is applied, the user's vestibular system is stimulated in the direction opposite to the rotational direction on the global coordinate system to match the visual-vestibular information.

Translation Manipulation. In the translation manipulation environment, we can apply two RDW techniques: a virtual path longer ($g_T > 1$) or shorter ($g_T < 1$) than the real path. In the VE, the user walks in a straight direction, and thus feels the acceleration in the backward direction. At this time, if a virtual path longer than the real path ($g_T > 1$) is provided to the user, a visual virtual acceleration larger than the vestibular acceleration will be felt. Therefore, we reduce visual-vestibular inconsistency by stimulating the same pitch direction as the acceleration direction on the global coordinate system. Conversely, when a shorter virtual path ($g_T < 1$) is applied, stimulation is performed in the opposite pitch rotation direction to accelerate the global coordinate system.

Walking Direction Manipulation. Users can experience two RDW techniques: left curvature ($g_C < 0$) or right curvature ($g_C > 0$) in the virtual path in the walking direction manipulation environment. The user moves along a virtual straight path, but it follows an arc path in a specific direction in the RE. To solve the visual-vestibular inconsistency that comes from this, we imposed GVS stimulation in the same yaw direction to match the visual-vestibular information on the global coordinate system in each case.

Head coordinate system and global coordinate system. The REVES system device provides vestibular stimulation in six rotation directions (right/left roll, front/rear pitch, right/left yaw) based on the user's head coordinate system by controlling the current direction. However, depending on the user's head orientation, the rotation direction of the head coordinate system may not match the rotation direction of the global coordinate system. For example, when the user is looking straight ahead, LDS imposes a stimulus in the roll rotation direction in the global coordinate system. However, if the user is looking at the floor, LDS gives a stimulus in the yaw rotation direction in the global coordinate system. Therefore, we tracked the user's head orientation using IMU and transformed the coordinates to stimulate the user in the desired direction on the global coordinate system.

4 EVALUATION

4.1 Participants

We recruited 11 participants (two females, age range 22–28, $M = 24.8$, $SD = 1.54$). Pregnant women and people with vestibular system history, sensitive skin, use of a pacemaker, history of epileptic seizures, childhood absence epilepsy or migraine, brain damage, and central nervous system or cardiovascular disorders were excluded. The participants had previous experience in wearing HMDs, had no symptoms of motion sickness with 3D games, and did not have any problems with normal gait.

4.2 Experiment Setup

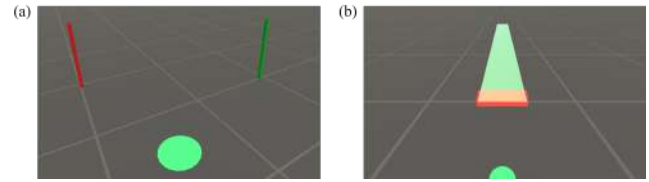


Figure 5: Experimental environments used in (a) E1 and (b) E2 and E3. In E1, the participant looked at the green bar and then rotated on the green plate to see the red bar. In E2 and E3, participants had to walk straight along the green road, starting from the red block.

Before the experiment, we checked whether the participants felt discomfort with electrical stimulation by pre-imposing the GVS. If the participant felt uncomfortable with the stimulations and was unable to proceed, we immediately terminated the experiment. The participants had 3 min to accustom to the VE. We used the DT and PSE measurement experiments proposed by Steinicke et al. [26] using a 2AFC task to determine PSE and DTs for all experiments. After each experimental session, the participants responded to a simulator sickness questionnaire (SSQ) to measure motion sickness [10]. The participant always wore an HMD with GVS while participating in the experiment. With the application of each gain, GVS was given to the participant in the REVES condition. In the control condition, the power to the GVS was disconnected; therefore, RDW was performed using only visual modulation without GVS. The experiment consisted of three parts (referred to as Experiment 1, 2, and 3) in the environment shown in the Figure 5. At least 30 min of resting time were provided between consecutive parts. All experiments and conditions were randomized.

Experiment 1: Detection threshold of REVES system measurement in rotation gain (E1). This experiment investigated the participant's ability to distinguish whether the physical rotation is smaller or greater than virtual rotation. First, the participant looked at the green bar for one second, which was generated in a random direction within a radius of 1.5 m around the participant. Thereafter, the randomized rotation gain was imposed on the participant, and the participant turned either left or right toward the red bar, with the angle of 90 degrees. They were instructed to rotate freely using both their body and head. After looking at the red bar for one second, the bars were removed, and the 2AFC panel was presented.

The participants had to determine whether the simulated virtual rotation was smaller or greater than the physical rotation, selecting the answers by clicking the UI button in VE using the trigger button of the controller. The green bar was regenerated randomly for the subsequent trial. We varied rotation gain between 0.6 (150-degree physical rotation resulted in a 90-degree virtual rotation) and 1.4 (64-degree physical rotation resulted in a 90-degree virtual rotation) in steps of 0.1. We tested each gain six times in a randomized order for REVES and control conditions.

Experiment 2: Detection threshold of REVES system measurement in translation gain (E2). This experiment investigated the ability to discriminate between virtual and physical straight, forward movements. After initializing, the participants moved a 5-m virtual distance along a straight green path. When the participants traveled 5 m along the path in the VE, the participants presented the 2AFC panel. They had to determine whether the virtual distance was longer or shorter than the physical distance and they selected the answer by clicking the UI button in the VE using the trigger button. After selecting an answer, the participants returned to the starting position marked with a green plate for the next experiment. To prevent the participants from estimating the distance using the plate, we temporarily removed the floor grid until the participants return to the starting point. We varied the translation gain between 0.76 and 1.24 in steps of 0.06, that is, the distance the participants had to walk varied between 3.8 m and 6.2 m in RE. We tested each gain five times in a randomized order for REVES and control conditions.

Experiment 3: Detection threshold of REVES system measurement in curvature gain (E3). This experiment analyzed the participant's ability to recognize curvature gains. To avoid participants noticing the curve when curvature gains were applied to the participants immediately after departure, we applied curvature gains after the participant walked a straight 1.5-m segment without any manipulation. After walking straight for 1.5 m, the participants moved a 5-m virtual distance to which curvature gain was applied: they walked a total of 6.5 m along the one-meter-wide green straight path in VE. The participants had to decide whether the physical path was curved to the left or the right. After answering, the participants returned to the green plate and proceeded with the following task. We removed the grid from the floor to prevent the participants from recognizing the curve based on the plate. We varied the rotation of the scene by 5, 10, 15, and 20 degrees per 5-m distance to either side, and this can be calculated as a curvature gain of $\pm\pi/180$, $\pm\pi/90$, $\pm\pi/60$, $\pm\pi/45$. We tested each gain five times in a randomized order for REVES and control conditions.

5 RESULTS AND DISCUSSION

5.1 Results

Results of E1. Figure 6 (a) shows the pooled results at rotation gain $g_R \in \{0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6\}$. On the x-axis of the graph is rotation gain g_R , and the y-axis is the probability that the participant answered that the virtual rotation was smaller than the physical rotation, and standard error is indicated. The solid line in the graph is the result of fitting each result value to

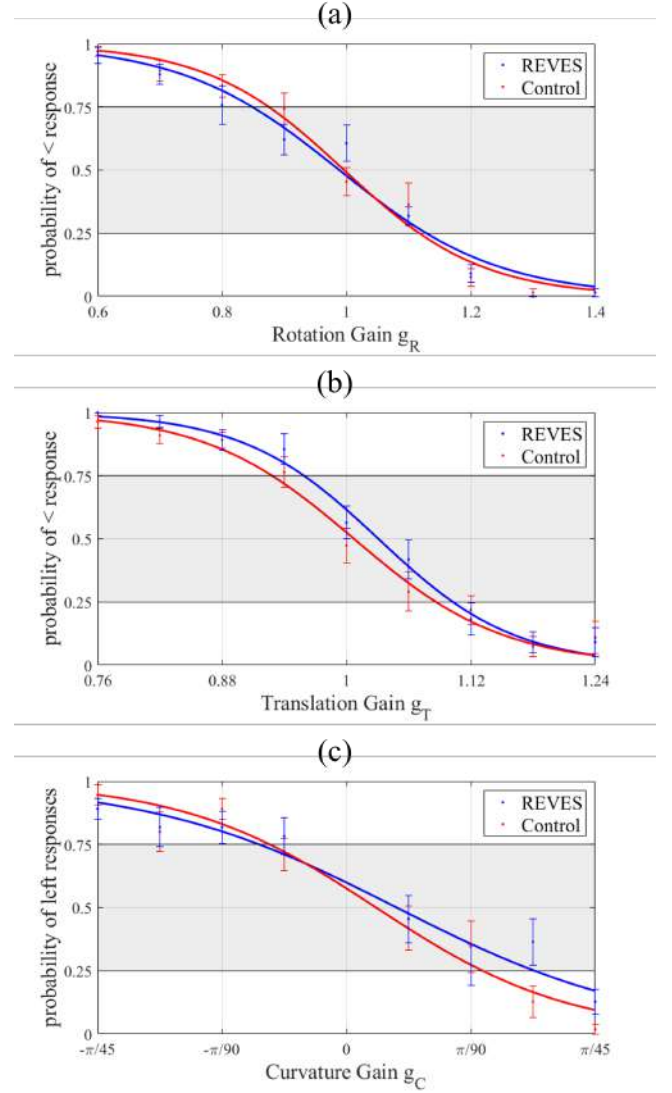


Figure 6: Pooled result of E1 (a), E2 (b), E3 (c). The results of the REVES system condition are shown in blue, and the control condition is shown in red. The solid line in the graph is the result of fitting each result value to the psychometric function. The 25% and 75% lines were marked on the graph, and the inner area was painted in the shade.

the psychometric function. The exact values of PSE, lower DT, and upper DT for each condition obtained from psychometric functions are summarized in Table 1. Results confirmed that the pooled DT range in the REVES condition was expanded by 16.225% compared to the control condition.

Results of E2. Figure 6 (b) shows the pooled results at translation gain $g_T \in \{0.76, 0.82, 0.88, 0.94, 1.00, 1.06, 1.12, 1.18, 1.24\}$. On the x-axis of the graph is translation gain g_T , and the y-axis is the probability that the participant answered that virtual translation

Table 1: Lower DT (LDT), upper DT (UDT) and PSE for REVES and control condition in E1 (g_R), E2 (g_T), E3 (g_C).

Conditions	E1 (g_R)			E2 (g_T)			E3 (g_C)		
	LDT	PSE	UDT	LDT	PSE	UDT	LDT	PSE	UDT
REVES	0.848	0.988	1.129	0.959	1.301	1.103	-0.025	0.014	0.053
Control	0.875	0.996	1.117	0.929	1.007	1.087	-0.021	0.008	0.038

was smaller than the physical translation, and standard error is shown. The exact values of PSE, lower DT, and upper DT of each condition obtained from psychometric function are summarized in Table 1. Results confirmed that the pooled DT range in the REVES condition was reduced by 8.571% compared to the control condition.

Results of E3. Figure 6 (c) shows the pooled result at curvature gain $g_C \in \{\pm\pi/180, \pm\pi/90, \pm\pi/60, \pm\pi/45\}$. In the x-axis of the graph is curvature gain g_C , and the y-axis is the probability that the participant answered that the physical path was curved to the left, and the standard error is shown. The exact values of PSE, lower DT, and upper DT for each condition obtained from psychometric functions are summarized in Table 1. Results confirmed that the pooled DT range in the REVES condition was expanded by 30.108% compared to the control condition.

Motion Sickness. We performed SSQ to measure motion sickness caused in each experimental condition, and the average result of SSQ total severity (SSQ TS) was 30.713 ($SD = 38.275$) in the control condition and 32.980 ($SD = 29.470$) in the REVES condition. The results were analyzed through a paired t-test but a significant change in motion sickness with or without the REVES system could not be confirmed ($t = -0.432$, $p = 0.669$). This result means that the REVES system did not significantly increase motion sickness compared to the control condition.

5.2 Discussion

Detection Threshold. The range of pooled DT measured in rotation manipulation (g_R) and walking direction manipulation (g_C) increased by 16.225% and 30.108%, respectively, and the range of redirection gain available to users was expanded without a change in the SSQ TS score. Therefore, under the REVES system, it was more challenging to notice the manipulation of the virtual environment without increased motion sickness. Therefore, the REVES system increased the range of rotation manipulation and walking direction manipulation through the four-pole vestibular stimulation, which showed the possibility of increased spatial expansion of RDW in each manipulation. In translation manipulation (g_T), the range of pooled DT was reduced to 8.571%. Unlike the other two manipulations, the direction of visual information modulation given to the user in translation manipulation is linear, not rotational. Therefore, it is assumed that the REVES system, which provides vestibular stimulation in the 3-axis rotation direction, did not have a significant effect on translation manipulation. Additional experiments are required to prove this. However, the DT range in the $g_T > 1$ region increased by 18.497%, shifting the entire graph to the right. This result means that the multi-directional vestibular stimulation of the REVES system has a substantial effect on the user's perception of translation manipulation, and more detailed

research on the case is required.

Point of Subjective Equality. The PSE values in rotation manipulation were 0.988 in the REVES condition and 0.996 in the control condition, and slightly increased when the REVES system was applied. This result suggests that virtual rotation was slightly overestimated in the REVES condition. The PSE values for translation manipulation and walking direction manipulation were 1.031 and 0.014 in the REVES condition and 1.007 and 0.008 in the control condition. In both cases, the PSE values increased in the REVES condition. This may imply that the REVES system caused an overestimation of the length of the virtual path in translation manipulation and caused an overestimation of the right curvature in walking direction manipulation. However, unintentional bias occurred in our walking direction manipulation experiment because the PSE value in the control condition increased compared to the PSE value of 0.002 in a previous study by Steinicke et al [27]. under the same condition. Therefore, the effect of the REVES system on overestimation in walking direction manipulation seems to require additional research.

6 CONCLUSION AND FUTURE WORKS

This study investigated the effect of the newly proposed REVES system on DT and PSE under each manipulation condition. In rotation manipulation and walking direction manipulation, the REVES system demonstrated the possibility of increasing spatial expansion by increasing the range of pooled DT of RDW. Although REVES did not expand the range of DT in translation manipulation, it was still possible to confirm interesting changes in PSE values. It is expected that the effect of the REVES system can be further strengthened by utilizing these interesting changes in DT and PSE in future studies.

In the future, we intend to further study the verification of REVES system in general VR situations. Furthermore, we plan to verify this system in a practical game environment using generalized RDW algorithms, such as steer-to-center and steer-to-orbit [19], outside the controlled laboratory environment. The hardware of the REVES system is a prototype, and there is plenty of room for improvement. In future research, we will improve the system performance by adding several additional sensor modules or new types of stimulation. Furthermore, we will enrich the study by recruiting more varied participants (e.g. gender, age).

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